InkDrop - Database Requirements Specification

1. Data Entities & Attributes

1.1. Users

- Purpose: Stores information for all system users across different roles (Customers, Print Shop Operators, Delivery Agents).
- Attributes:
    - User_ID (Primary Key, Auto Increment)
    - Full_Name (VARCHAR, Required)
    - Email (VARCHAR, Unique, Required)
    - Password_Hash (VARCHAR, Required)
    - Phone_Number (VARCHAR, Required)
    - Role (ENUM: 'Customer', 'Operator', 'Delivery_Agent', Required)
    - Created_At (TIMESTAMP, Default CURRENT_TIMESTAMP)

1.2. Documents

- Purpose: Stores metadata for files uploaded by customers for printing.
- Attributes:
    - Document_ID (Primary Key, Auto Increment)
    - Customer_ID (Foreign Key -> Users.User_ID)
    - File_Name (VARCHAR, Required)
    - File_Path (VARCHAR / Secure URL, Required)
    - File_Size_MB (DECIMAL, Required)
    - Page_Count (INT, Required)
    - Uploaded_At (TIMESTAMP)

1.3. Print_Configurations

- Purpose: Lookup table storing available print options and their corresponding base prices per page.
- Attributes:
    - Config_ID (Primary Key, Auto Increment)
    - Paper_Size (ENUM: 'A4', 'A3', 'A5')
    - Color_Mode (ENUM: 'Black_White', 'Color')
    - Layout (ENUM: 'Single_Sided', 'Double_Sided')
    - Binding_Type (ENUM: 'None', 'Spiral', 'Thermal', 'Hardcover')
    - Price_Per_Page (DECIMAL(10,2), Required)

1.4. Orders

- Purpose: Central entity representing each print-and-delivery transaction.
- Attributes:
    - Order_ID (Primary Key, Auto Increment)
    - Customer_ID (Foreign Key -> Users.User_ID)
    - Operator_ID (Foreign Key -> Users.User_ID, Nullable)
    - Delivery_Agent_ID (Foreign Key -> Users.User_ID, Nullable)
    - Total_Amount (DECIMAL(10,2), Required)
    - Delivery_Address (TEXT, Required)
    - Delivery_Area_Fee (DECIMAL(10,2), Required)
    - Order_Status (ENUM: 'Pending', 'Printing', 'Ready_For_Pickup', 'Out_For_Delivery', 'Delivered', 'Cancelled')
    - Created_At (TIMESTAMP)

1.5. Order_Items

- Purpose: Bridges orders with uploaded documents and their specific print configurations (resolves Many-to-Many relationship between Orders and Documents).
- Attributes:
    - Item_ID (Primary Key, Auto Increment)
    - Order_ID (Foreign Key -> Orders.Order_ID)
    - Document_ID (Foreign Key -> Documents.Document_ID)
    - Config_ID (Foreign Key -> Print_Configurations.Config_ID)
    - Quantity_Copies (INT, Default 1)
    - Subtotal_Price (DECIMAL(10,2), Required)

1.6. Payments

- Purpose: Records transaction and payment details for each order.
- Attributes:
    - Payment_ID (Primary Key, Auto Increment)
    - Order_ID (Foreign Key -> Orders.Order_ID, Unique)
    - Payment_Method (ENUM: 'Cash_On_Delivery', 'Online_Card')
    - Payment_Status (ENUM: 'Pending', 'Completed', 'Failed', 'Refunded')
    - Transaction_Reference (VARCHAR, Nullable)
    - Paid_At (TIMESTAMP, Nullable)

2. Business Rules & Data Constraints

1. Unique Identity Constraint: The Email field in the Users entity must be unique across the system to prevent duplicate user registration.
2. Referential Integrity:

    - An order (Orders) cannot be created without a valid Customer_ID existing in the Users table.
    - Account deletion enforces ON DELETE RESTRICT to preserve historical financial and order logs.
3. Domain Constraints:
    - Page_Count and Quantity_Copies must be strictly positive integers (greater than 0).
    - Monetary amounts (Total_Amount, Price_Per_Page, Subtotal_Price) must be non-negative values (greater than or equal to 0.00).
4. Role Authorization Rules:
    - Operator_ID in Orders can only reference a User_ID whose Role is 'Operator'.
    - Delivery_Agent_ID can only reference a User_ID whose Role is 'Delivery_Agent'.
5. Dynamic Pricing Calculation Rule: The item subtotal (Subtotal_Price) is calculated dynamically using the formula: Subtotal_Price = (Page_Count * Price_Per_Page) * Quantity_Copies

3. Key Database Queries & Operations

- Q1 (Customer View): Retrieve all orders placed by a specific customer along with real-time status and total cost.
- Q2 (Operator Dashboard): Fetch all orders with 'Pending' status alongside document download links to initiate printing.
- Q3 (Delivery Assignment): Update Delivery_Agent_ID for an order and change its status to 'Out_For_Delivery'.
- Q4 (Sales & Revenue Analytics): Aggregate daily and monthly revenue broken down by popular print configurations.